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INTERNSHIP TASK REPORT

TASK 01

SPECIFICATION AND COA READING

Documentary Quality Assessment of Sucrose Extrapure

Product	Sucrose Extrapure 500 GM
Manufacturer	Actylis
Batch number	961820225CZ
Internship	Pharmaceutical Quality Control
Submitted by	JEET SENGUPTA
Submission date	04/11/2026

Educational documentary review based on publicly available manufacturer documentation

1. Introduction

Quality Control (QC) is an essential part of manufacturing and supply operations. It involves evaluating materials and products against predefined quality requirements. A specification describes the quality attributes and acceptance criteria that a material is expected to meet. A Certificate of Analysis (COA) records the results reported for a particular batch against specified tests.

Reviewing a COA against its corresponding specification helps assess whether the reported batch results meet the stated requirements. This report presents a documentary review of a manufacturer-issued COA for Sucrose Extrapure. The exercise focuses on interpreting acceptance criteria, comparing reported results, and explaining the limitations of OOS and OOT assessments.

2. Objectives

- To understand the purpose of material specifications and Certificates of Analysis.
- To examine the manufacturer-provided COA for Sucrose Extrapure.
- To compare reported batch results with the corresponding acceptance criteria.
- To assess the reported results and consider OOS/OOT observations based on the available documentation.

3. Product and Batch Information

Parameter	Details
Product name	Sucrose Extrapure 500 GM
Manufacturer	Actylis
Product / SAP code	11746SG500
Batch number	961820225CZ
Grade	Extrapure
CAS number	57-50-1
Molecular formula	$C_{12}H_{22}O_{11}$
Molecular weight	342.3
Manufacturing date	March 2026
Expiry date	February 2031

Sucrose is a disaccharide composed of glucose and fructose units. Its molecular formula is $C_{12}H_{22}O_{11}$. The selected material is identified by the manufacturer as Extrapure grade. It is not described in this report as USP or NF grade.

4. Source Document and Methodology

The primary source used for this exercise is the manufacturer-provided, batch-specific Certificate of Analysis for Sucrose Extrapure 500 GM, SAP code 11746SG500, batch number 961820225CZ. The acceptance criteria and results reproduced in this report are taken from the specification and result columns shown in that COA.

The product name, product code and batch identification were recorded from the COA.
The listed test parameters and their acceptance criteria were identified.
Each reported result was compared with its corresponding criterion.
The documentary assessment was classified as Pass where the reported result met the stated requirement.
OOS and OOT considerations were evaluated using only the information available in the selected COA.

No physical sample was obtained or tested. No independent laboratory measurements or verification of the manufacturer's analytical work were performed.

5. Results and Discussion

The following table summarises the six test parameters shown in the selected COA.

No.	Test parameter	Specification	COA result	Assessment
1	Description	Colourless crystals/crystalline powder	Colourless crystalline powder	Pass
2	Solubility	10% solution in water should be clear	Complies	Pass
3	Specific optical rotation	+66.4° to +66.6°	+66.5°	Pass
4	Residue on ignition (as SO_4)	Maximum 0.05%	0.02%	Pass
5	Heavy metal (as Pb)	Maximum 0.0005%	Less than 0.0005%	Pass
6	Reducing sugars	Passes test	Complies	Pass

5.1 Description

The specification describes the material as colourless crystals or crystalline powder. The reported result is colourless crystalline powder, which is consistent with the stated description. Assessment: Pass.

5.2 Solubility

The specification requires a clear 10% solution in water. The COA reports “Complies”. This is recorded as meeting the stated qualitative requirement. Assessment: Pass.

5.3 Specific optical rotation

The specified range is +66.4° to +66.6°, and the reported result is +66.5°. The result lies within the stated range. Assessment: Pass.

5.4 Residue on ignition (as SO_4)

The maximum permitted value is 0.05%, while the reported result is 0.02%. Since 0.02% is below 0.05%, the result meets the stated limit. Assessment: Pass.

5.5 Heavy metal (as Pb)

The maximum specified value is 0.0005%, and the reported result is less than 0.0005%. The result therefore meets the stated maximum limit. The notation does not mean that heavy metals are absent; it means the reported value is below the stated limit. Assessment: Pass.

5.6 Reducing sugars

The specification requires the material to pass the test, and the COA reports “Complies”. The reported qualitative result meets the stated requirement. No numerical concentration is provided in the COA. Assessment: Pass.

6. Out-of-Specification (OOS) Assessment

An out-of-specification (OOS) result is a reported result that falls outside the applicable acceptance criteria. Based on the six parameters reproduced from the selected COA, no OOS result was identified. Each numerical result is within its stated limit or range, and the qualitative tests are reported as complying with their requirements.

7. Out-of-Trend (OOT) Assessment

An out-of-trend (OOT) result is a result that may remain within specification but differs unexpectedly from an established historical pattern. Only one batch-specific COA was available for this exercise. Historical results, control charts or established trend limits were not available. Therefore, an OOT assessment cannot be conclusively performed from the information reviewed. This should not be interpreted as proof that no trend deviation exists.

8. Quality Assessment and Documentary Disposition

Based on the manufacturer-reported information reproduced in the COA, all six listed test results meet their corresponding stated acceptance criteria. No OOS result was identified. OOT assessment remains inconclusive because historical batch data were unavailable.

Documentary conclusion: the batch conforms to the listed criteria based on the available COA. This conclusion is limited to the document reviewed. It is not an independent confirmation of the material's quality, a receiving-site acceptance decision, or formal batch-release authorisation. Any actual material release would be subject to the receiving organisation's approved procedures and applicable requirements.

9. Conclusion

The manufacturer-provided COA for Sucrose Extrapure, batch number 961820225CZ, was reviewed against the acceptance criteria displayed in the same document. Six parameters — description, solubility, specific optical rotation, residue on ignition, heavy metal and reducing sugars were assessed.

All six reported results met their corresponding stated criteria.
No OOS result was identified.
An OOT conclusion could not be reached because historical batch data were not available.

This exercise provided practice in reading a COA, interpreting specification limits, comparing analytical results and documenting the limitations of a quality assessment.

10. References

Actylis. Certificate of Analysis: Sucrose Extrapure 500 GM, SAP code 11746SG500, batch number 961820225CZ. Manufacturer-issued electronic COA. Available at: https://www.actylislab.com/product_certificate_of_analysis?itemcode=11746SG500 (accessed 3 November 2026).

11. Disclosure and Limitations

This report is an educational case study prepared using publicly available manufacturer documentation. Product details, specifications and reported results were transcribed from the manufacturer's batch-specific COA. No physical sample was obtained or tested, no independent analytical measurements were performed, and the manufacturer's results were not independently verified. The selected product is identified as Extrapure grade and is not represented here as USP/NF pharmaceutical grade. The review does not constitute formal batch-release authorisation.

Appendix A — Original Manufacturer COA

Actylis Certificate of Analysis for Sucrose Extrapure, SAP code 11746SG500, batch number 961820225CZ

Source: https://www.actylislab.com/product_certificate_of_analysis?itemcode=11746SG500

actylis

Certificate of Analysis

Product Name	Sucrose extrapure 500GM		
SAP Code	11746SG500	**Batch No**	961820225CZ
Grade	Extrapure	**IR No**	9618202
CASR No.	57-50-1	**QC Ref. No**	119089
Molecular Formula	C12H22O11	**Manufacture Date**	March 2026
Molecular Weight	342.3	**Expiry Date**	February 2031

SR.NO.	TEST	RESULT	SPECIFICATION	METHOD REFERENCE
1	Description	Colourless crystalline powder.	Colourless crystals/crystalline powder	
2	Solubility	Complies	10% solution in water should be clear.	
3	Specific rotation([a]20°/D,c=5,H2O)	+66.5°	+66.4° to +66.6°	
4	Residue on ignition(as SO4)	0.02%	0.05% max.	
5	Heavy metal(as Pb)	Less than 0.0005%	0.0005% max.	
6	Reducing sugars (C6H12O6)	Complies	Passes test	

Remarks: THE CAPTIONED BATCH MATERIAL COMPLIES WITH THE PRESCRIBED QUALITY OF THE ABOVE SPECIFICATION.

Actylis provides this analytical report in good faith and to the best of its knowledge. It should not be substituted for the users own raw material quality control.

Tested By
(Chemist - Quality Control)

Approved By
(Head - Quality Control)

Computer Generated Report - Does not require Signature.

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Certificate of Analysis

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